Diana Kostadinova

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Angular | Kotlin / Java Spring Boot | PostgreSQL | AI/Ml Concepts

Education

- **Faculty of Computer Science and Engineering, UKIM, Skopje** Skopje, N. Macedonia
  Bachelor of Software Engineering and Information Systems; GPA: 8.5/10.0 *Oct. 2024 – June. 2028*

Experience

- **Sorsix, Skopje (Internship)** Skopje, N. Macedonia
  Software Engineer Intern; Angular, Kotlin Spring Boot, PostgreSQL *Mar 2026 – Present*
  Contributed to the development of full-stack web applications within a production-oriented environment, working across both **backend** and **frontend** components. Gained hands-on experience in building and maintaining scalable features, writing **unit tests**, supporting database-driven functionality, and working with **deployment workflows**. Strengthened understanding of backend development principles, including **data modeling**, **persistence**, and ensuring **consistency** across services.

  On the frontend, worked with modern **reactive patterns** to implement responsive UI components. Applied structured software engineering practices, including **clean architecture** and common **design patterns**, while actively participating in resolving real development issues. Delivered a functional **flashcards application** as a final project built on the same technology stack.

Projects

- **LabelyAI** : Built a multi-domain AI annotation platform that reduces manual labeling effort by up to **70%** via prompt-driven segmentation and human-in-the-loop refinement. Uses **SAM3** for foundation segmentation with domain-adaptive pipelines for **medical imaging**, **industrial defect detection (PatchCore)**, and **logo recognition (ResNet18)**. Backend built with **FastAPI**, **Spring Boot**, **Cloudflare R2**, and **PostgreSQL** to support automated mask generation, anomaly scoring, and large-scale dataset production.
- **Skope**: Built a tourism mobile application with **React Native** that identifies landmarks in real time. Leveraged **Python**, **FastAPI**, and **Roboflow** with **YOLOv8** for object detection, managing extensive datasets that were cleaned, annotated, and optimized for model training. The project combined backend AI processing with a user-friendly front-end interface, delivering accurate, real-time landmark recognition for tourists.
- **KonKat**: A social media platform for professional networking, similar to LinkedIn but focused on direct, community-driven connection. Built with **Angular** on the front-end, **Kotlin** and **Spring Boot** on the back-end, and **PostgreSQL** for data persistence—covering user profiles, posts, follows, and real-time interaction features.
- **DiscoverMacedonia**: Developed a dynamic web application using **React** and **Firebase**, integrating multiple APIs and large language models (LLMs) via the **OpenAI API**. Gained hands-on experience with **Hugging Face** AI models and AI agents, designing interactive features that deliver advanced AI-powered experiences.

Publications

- **Hybrid Embedding and Classification Framework for Landmark Recognition with Limited Data (CiiT 2026)**: Designed and implemented a hybrid metric-classification model for few-shot landmark recognition using **Python** and **PyTorch**. Combined metric learning with a classification branch and aggressive data augmentation to handle extremely limited datasets (∼30 images per class). Achieved robust embeddings and ∼75% classification accuracy across Macedonian landmarks, demonstrating the effectiveness of hybrid few-shot learning in real-world, low-data scenarios.
- **Prompt-Based Annotation Framework for Large-Scale Visual Datasets (CiiT 2026)**: Co-authored a research paper on an AI-assisted visual annotation framework aimed at reducing manual labeling effort in large-scale datasets through domain-adaptive model specialization. The study extends the **SAM3** foundation model by introducing task-specific adaptations for medical imaging, industrial inspection, and logo recognition, demonstrating consistent improvements in segmentation fidelity and downstream classification accuracy compared to the base model. The system is designed around a scalable distributed architecture enabling automated mask generation, anomaly estimation, and efficient dataset curation across heterogeneous visual domains.